

## Review

# Adoption determinants and policy tools for residential green stormwater infrastructure: A review synthesizing differences and commonalities among lot-level practices

Hazem Ahmed <sup>\*,1</sup>, Dawn Cassandra Parker, Michael Drescher

School of Planning, University of Waterloo, 200 University Ave. West, Waterloo, ON N2L 3G1, Canada

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## ABSTRACT

Climate change adaptation in intensifying urban environments benefit from green stormwater infrastructure (GSI) investments on private residential yards. Nevertheless, planners are challenged to devise policy tools to mesh such a decentralized GSI approach with current land-use and social systems. Prior research has addressed the multi-scalar socio-economic barriers hindering household uptake, including technical and governance considerations. However, the interconnectedness of these factors has not received much attention. Additionally, studies into GSI often analyze a specific GSI but then overgeneralize their conclusions about adoption determinants for a range of GSI practices. Our review aims to refine previously ambiguous generalizations by comparing adoption factors for three distinct GSI practices: rain gardens, rainwater harvesting, and permeable pavers. We systematically identified 1753 studies and conducted a full content analysis of 56 studies, defining 17 explanatory factors and discussing their independent influences on the three GSI practices. Our results suggest that adoption factors vary between GSI practices, highlighting the need for practice-specific policy tools. Using the Fogg Behavioral Model as a qualitative framework, we illustrate and synthesize the interplay between motivational and ability factors and propose potential targeted policy interventions for each GSI practice. Evidence from different contexts on the three practices suggests that neither stimulative policy instruments (e.g., providing cost subsidies) nor restrictive tools (e.g., enforcing minimal infiltration rates) are efficient in isolation. Our results can guide scholars, decision-makers, and professionals to craft practice-specific integrated policy packages, accounting for socioeconomic factors to achieve transformative GSI uptake.

## 1. Introduction

### 1.1. Green stormwater infrastructure on private residential yards: challenges and opportunities

Stormwater management systems are encountering a transition in their ideological and technical forms. Conventional infrastructure planning primarily relies on engineering conceptions (Furlong et al., 2016), which focuses on functional effectiveness, but overlooks environmental and social dynamics (Salet et al., 2013). These limitations have motivated the relatively new decentralization approach, which replaces the principle of investing in "economies of unit scale" with "economies of numbers" (Bell, 2015). In other words, large grey infrastructural facilities should be complemented by large numbers of

small-scale innovative technologies (Dahlgren et al., 2013), including green stormwater infrastructure (GSI) at the parcel level, such as rain gardens or rainwater harvesting (Walsh et al., 2022).

Such collective action of decentralized GSI interventions at the residential parcel-level could improve the resilience of stormwater management systems. Residential yards are an example of fine-scale, unbuilt spaces, whose importance stems from their frequent recurrence throughout the city core and fringes (Cameron et al., 2012; Forman, 2008; Goddard et al., 2010; Mathieu et al., 2007). Further, within urban contexts, private properties generate more than 60% of urban runoff (Maragno et al., 2018). Therefore, where low-density housing, such as single detached homes, is dominant, decentralized GSI installations on residential yards could alleviate pressure on grey infrastructure and minimize required retrofits of existing drainage or storage systems

\* Corresponding author.

E-mail addresses: [Hazem.Ahmed@uwaterloo.ca](mailto:Hazem.Ahmed@uwaterloo.ca) (H. Ahmed), [dcparker@uwaterloo.ca](mailto:dcparker@uwaterloo.ca) (D.C. Parker), [mdrescher@uwaterloo.ca](mailto:mdrescher@uwaterloo.ca) (M. Drescher).

<sup>1</sup> Corresponding author at all stages of refereeing, publication, and post-publication.

(Mayer et al., 2012).

Although GSI seems integral to sustainable urban water management systems, its decentralized operationalization is subject to complex challenges arising from synchronizing natural, engineered, and social systems, which hinder GSI uptake at institutional and household scales. Stormwater management systems are no longer only underground engineered systems but incorporate private open spaces. In this context, alongside stormwater engineers and other infrastructure professionals, residents can contribute to stormwater management systems (Beery, 2018; Geldof, 1995; Wilfong et al., 2024), and their individual yard management behaviors could collectively alter an urban water system at large (Coleman et al., 2018; Maragno et al., 2018; Söderberg and Åberg, 2002; Sun and Hall, 2016). Therefore, a sustained collaboration between multiple stakeholders is needed to design, construct, and operate decentralized GSI practices (Montalto et al., 2013; Schiffman et al., 2017). Planners can facilitate the collaborative action required across different disciplines needed to develop strategies and tools that consider long-term climate change objectives (Hurlimann and Wilson, 2018). However, such facilitation requires an in-depth understanding of the determinants and barriers to GSI adoption by resident households.

## 1.2. Review scope and objectives

This review aims to comprehensively investigate the social determinants and barriers to GSI adoption within residential neighborhoods, focusing on the household scale. GSI practices vary in their requirements and benefits to residential households (Taguchi et al., 2020). Therefore, we hypothesize that adoption determinants vary from one GSI to another. Furthermore, we seek to investigate how adoption determinants may interact with household characteristics to shed light on the complexity of household adoption dynamics.

With the growing interest in GSI implementation, literature addressing barriers to effective GSI implementation is also increasing. As this interdisciplinary literature has evolved, it has mirrored the ideological and technical transformation in stormwater management system practices. Initially, it focused on engineering aspects, but recently it has shifted towards studying social perspectives. Nevertheless, GSI scholars and professionals acknowledge that households' yard management decisions remain largely detached from current municipal stormwater plans and policies, raising decentralized GSI reliability concerns (National Research Council, 2009; Sun et al., 2024). We argue that generalized conclusions about factors influencing GSI adoption, which we discuss below, may lead to misleading recommendations regarding the factors influencing households' uptake for each GSI type. This lack of specificity is insufficient for evidence-based policy regarding planning for household-level uptake of GSI.

Thus, this review aims to address two critical gaps in the literature. First, there is a lack of studies that relate household uptake determinants to specific GSI practices. While there are studies that explicitly address one practice (Irungu et al., 2024; Kelemewerk Mekuria et al., 2020), many of them assert general conclusions about adoption determinants of green infrastructure while studying only particular GSI applications (Baptiste et al., 2015; Maeda et al., 2018; Tanaka et al., 2022; Venkataramanan et al., 2019, 2020). We acknowledge a recently published review that sought to identify socio-economic factors influencing the adoption of GSI practices; however, it included only literature within the US context (Abuismail et al., 2024), which highlights the need for a more global perspective. Further, similar to previous works, Abuismail et al. (2024) addressed GSI generally in their discussion and conclusions, lacking specificity for individual GSI practices. We also contend that, while current research identifies various GSI adoption determinants, scholars often discuss each in isolation (e.g., Venkataramanan et al., 2020), overlooking their interconnectedness.

Our review addresses these limitations by collecting evidence from diverse geographical and cultural contexts on adoption determinants for different GSI practices. More specifically, we investigated three GSI

practices: (1) rain gardens, (2) rainwater tanks or barrels, and (3) permeable pavers. To our knowledge, this is the first systematic review that compares three different GSI adoption determinants while comprehensively reporting the diversity of the dependent variables addressed in the literature and reporting findings on each practice separately. Furthermore, we have developed a qualitative framework that synthesizes the complexity of households' adoption decisions, based on the Fogg behavioral model (Fogg, 2009, 2020). Our framework illustrates the interplay between adoption determinants, providing a practical tool for understanding the dynamics of household GSI adoption.

The overarching goal of our review is to provide a comprehensive understanding of the GSI adoption determinants based on a comparative analysis that reveals the differences and commonalities between the three GSI practices and synthesizes their interrelationships. This systematic understanding of adoption determinants and their interrelationships will interest scholars and professional environmental planners, stormwater managers, landscape architects, and non-profit organizations studying or designing GSI policies, programs, and interventions within residential neighborhoods.

## 2. Methods

### 2.1. Databases and search strategies

We systematically searched and reviewed (Grant and Booth, 2009) scientific and grey literature on GSI adoption, including reports on stormwater programs and dissertations. We used Scopus and Google Scholar to identify relevant research published in English from the default starting date of Scopus (1990) through 2024. The search string included synonyms for GSI (Supplementary data 1) informed by relevant reviews and literature on the applications of green infrastructure, terminologies, and definitions (Bartessaghi-Koc et al., 2019; Benedict and McMahon, 2006; Fletcher et al., 2015; Young et al., 2014). We also consulted stormwater management experts to verify the various commonly used terms. We sought to restrict the search terms to address social perspectives and excluded technical efficacy and engineering studies (Supplementary data 1). For Google Scholar, we considered the first 300 results (Haddaway et al., 2015). Finally, we hand-searched references cited in the included studies.

### 2.2. Screening and selection process

This review is focused on green infrastructure to manage stormwater and flooding. Therefore, we screened out literature that covers other green infrastructure functions, such as mitigating or adapting to urban heat island effects or drought. However, we included studies that simultaneously address multiple green infrastructure benefits, while our results and discussion only synthesize findings related to stormwater management. We confined data collection to evidence on each GSI practice separately. Hence, we excluded any generalized syntheses or conclusions addressing GSI without specifying the exact practice.

We examined the adoption determinants of three GSI practices on private residential yards that are intentionally installed by households to address stormwater, including rain gardens (including residential natural landscaping and bioretention), rain barrels, and permeable pavers. We did not include trees as a reviewed GSI practices. Although trees are scientifically proven to contribute to stormwater management (Berland et al., 2017; Zölch et al., 2017), evidence asserts that households' awareness of this contribution is minimal. Instead, households primarily recognize tree benefits in terms of shading, cooling, and aesthetics (Conway et al., 2022). Furthermore, this review focused on social aspects, including uncertainties of adopting innovative GSI practices, and these do not apply to tree planting in residential settings. Innovative GSI here refers to landscape interventions that are not yet widely accepted by communities as conventional practices (Rogers, 2003) to manage

rainwater or flooding issues on their yards. We also excluded underground interventions, such as soak-aways and French drains, as our objective was to focus on practices that may interfere with households' daily life constraints, such as their need for more open space and aesthetics.

### 2.3. Fogg model as a theoretical lens for GSI adoptions

Investigating GSI adoption determinants requires understanding the decision-making mechanism of households (Gao et al., 2018). Various behavioral models have been devised to define determinants of behaviors (Dietz et al., 2005). There are classic normative behavior models, such as the Value-Believe-Norm theory (Stern et al., 1999) and Norm Activation Theory (Schwartz, 1970, 1973, 1977), which focuses on the role of extrinsic influences (e.g., social or moral norms) on an individual's altruistic behaviors (Turaga et al., 2010). In contrast, cognitive models such as the Theory of Reasoned Action (TRA) (Ajzen and Fishbein, 1980) primarily address the role of intrinsic factors on rationalized behaviors. The Theory of Planned Behavior (Ajzen, 1991, 2020), the extended version of the TRA, could be considered a hybrid model that integrates cognitive and normative approaches. It primarily addresses the intrinsic influences of the individual (e.g., personal beliefs) while accounting for, yet with limitations, normative beliefs and behavior-constraining factors (i.e., perceived behavioral control) that are considered extrinsic elements.

Our background review of studies addressing barriers to residential GSI adoption suggests that the use of the Theory of Planned Behavior to investigate and explain the socio-psychological factors influencing household adoption is academically prevalent (Brown et al., 2021; Drescher and Sinasac, 2021; Qi and Barclay, 2022; Shaw et al., 2011). Other studies follow Rogers's (2003) Diffusion of Innovation model to synthesize the stages of innovation adoption for households (Brown et al., 2021; Cote and Wolfe, 2014; López-Maciel et al., 2023; Shin and McCann, 2018). While the Diffusion of Innovation model is not formally a behavioral change model, it suggests a decision-making process that incorporates the role of individual characteristics and perceptions (i.e., intrinsic), and the influence of social dynamics through different communication channels (i.e., extrinsic) (Rogers, 2003).

The Fogg Behavioral model (Fogg, 2009) is a comprehensive framework developed to synthesize the determinants that influence human behavioral change. It stems from the use of technology as a persuasive tool to induce behavioral changes. Soon after its development, Fogg model applications extended to many other scientific disciplines, including the study of pro-environmental behaviors (Cheek et al., 2015; Pedersen et al., 2022; Wongtada et al., 2020). This behavioral

model builds on other classical models and comprehensively incorporates the role of intrinsic and extrinsic factors when investigating an individual's intentions and potential consequent actions. It is built on the principle that behavioral change results from three converging key elements: motivation, ability, and prompts (triggers or nudges) (Fig. 1). For example, to induce a pro-environmental behavior, such as installing a rain barrel on one's private yard, a "gardener" homeowner's desire might rise when learning about the benefits of collecting rainwater for watering purposes, especially in drought contexts (i.e., motivation). Easy-to-follow installation steps and know-how information on rain barrel maintenance could decrease action complexity (i.e., ability), and by the end of the winter season, timely reminders could signal them to buy and install the barrel (i.e., prompts).

The advantage of the Fogg model lies in its straightforward language and intuitive constructs, making it accessible to scholars and professionals who have little or no background in socio-psychological and behavioral science. For instance, the Fogg model suggests that intrinsic and extrinsic factors, including values, norms, beliefs, attitudes, and knowledge, could be aggregated under "motivations". Therefore, researchers are not required to know the interconnectedness or distinguish the differences between these concepts. Like one of the core elements of the Theory of Planned Behavior (i.e., perceived behavioral control), the Fogg model considers the complexity of the behavior in relevance to personal abilities and resources available to implement and use the adopted practice (e.g., physical effort). The Fogg model addresses a claimed limitation of the Theory of Planned Behavior, namely that it overlooks the impact of subtle yet repetitive triggers on decision-making (Sniehotta et al., 2014). A recent systematic review demonstrates that the "nudging" approach has a positive influence on inducing pro-environmental behaviors (Wee et al., 2021). In this context, we apply the Fogg model as a theoretical foundation to synthesize household-related factors while accounting for the influence of external prompts or nudges, such as policy tools.

## 3. Results

### 3.1. Characteristics of the reviewed literature

To test our hypothesis about the variability of adoption determinants, we collected qualitative and quantitative evidence from the reviewed literature. The majority of studies were quantitative, with only five studies presenting qualitative inferences, such as post-adoption field visits or reporting the majority of survey responses without testing their statistical significance. We also analyzed the type of dependent variables to comprehensively understand the impact of each factor on the

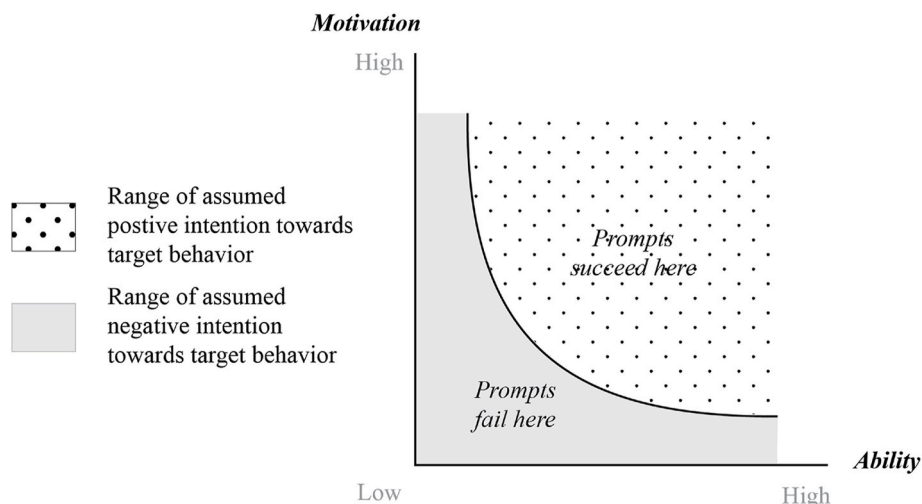


Fig. 1. Fogg behavioral model. Adapted from Fogg (2009, 2020).

adoption of households. We classified dependent variables into three groups: (1) willingness to adopt GSI, (2) willingness to pay for GSI installation, and (3) actual adoption of GSI. We systematically identified 1753 studies and conducted a full content analysis of 56. A flow diagram illustrating the search and article screening process following PRISMA (Page et al., 2021) is included in [Supplementary data 2](#). We also included a brief descriptive analysis of our search results in [Supplementary data 3](#).

### 3.2. Identified GSI adoption determinants

We identified 17 explanatory factors influencing households' decision to adopt GSI on their private yards. Some of these factors are practice-specific. For instance, one factor that is only related to rain barrels is the distance between the residence and the location where a rain barrel could be purchased. Furthermore, following the Fogg Behavioral Model (Fogg, 2009) we categorized each explanatory factor as a motivation or an ability (Table 1). All behavioral factors that might limit a resident's capacity to adopt GSI are grouped into "ability" factors, including physical, monetary, and technical information requirements and the autonomy of yard management. In total, we identified five "ability" groups of independent variables, including household traits (e. g. income, level of know-how, autonomy), characteristics relevant to the GSI practice (cost, maintenance demand), and the spatial or structural attributes of the property. Motivational factors include demographic variables, environmental worldviews and behaviors, by-law obligations and trust in municipal programs, and other socio-cognitive factors related to residential GSI adoption. The following sections (3.3, 3.4, and 3.5) include the detailed findings on all adoption determinants for each of the three GSI practices.

### 3.3. Adoption determinants for rain gardens

Of the 14 studies that investigated determinants influencing household decisions to adopt a rain garden on their yard, more than half (57%) measured residents' intentions as a dependent variable to adopt ( $n = 8$ ) or to pay for implementation ( $n = 1$ ). The remaining studies (36%) analyzed the actual adoption of rain gardens ( $n = 5$ ) (Table 2, Fig. 2). While Table 2 summarizes the associations and their relevant citations, the Sankey diagram does not reflect the direction and strength of the association between the independent and dependent variables; it reports how many studies analyzed these variable pairs.

#### 3.3.1. Motivational factors for rain garden adoption

**3.3.1.1. Sociodemographic attributes.** Age, the most frequently mentioned socio-demographic variable, was consistently reported to have a negative association with all dependent variables, including actual adoption or other intentional behaviors. Therefore, overall, the results assert that elderly residents (sometimes defined as  $> 60$  years) are less likely to consider rain gardens on their private yards (Ren et al., 2020; Shin and McCann, 2018; Ureta et al., 2021) and have a lower willingness to pay for them (Newburn and Alberini, 2016). The two

**Table 2**

Adoption determinants for rain gardens and their associations with actual adoption, willingness to adopt, or pay.

| Type of influence | Independent variable                               | Association from literature                                                                                                                                                             |                                               |                                                                                           |
|-------------------|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|-------------------------------------------------------------------------------------------|
|                   |                                                    | Positive                                                                                                                                                                                | Non-Significant                               | Negative                                                                                  |
| Motivation        | Age                                                | –                                                                                                                                                                                       | –                                             | (Newburn and Alberini, 2016; Ren et al., 2020; Shin and McCann, 2018; Ureta et al., 2021) |
|                   | Education level                                    | –                                                                                                                                                                                       | Newburn and Alberini (2016)                   | –                                                                                         |
|                   | Gender (self-identified as female)                 | (Peng, 2010; Shin and McCann, 2018).                                                                                                                                                    | Baptiste et al. (2015)                        | –                                                                                         |
|                   | Social norms                                       | (Belan and Nenn, 2010; Shaw et al., 2011; Ureta et al., 2021)                                                                                                                           |                                               | –                                                                                         |
|                   | Aesthetics                                         | Conway et al. (2022)                                                                                                                                                                    | (Baptiste et al., 2015; Newburn et al., 2013) | Belan and Nenn (2010)                                                                     |
|                   | By-law enforcement                                 | Woodward et al. (2009)                                                                                                                                                                  | –                                             | –                                                                                         |
|                   | Trust in municipal programs                        | Belan and Nenn (2010)                                                                                                                                                                   | –                                             | –                                                                                         |
|                   | Direct benefits to household (flooding mitigation) | (Ren et al., 2020; Ureta et al., 2021)                                                                                                                                                  | (Brown et al., 2016; Conway et al., 2022)     | Woodward et al. (2009)                                                                    |
|                   | Environmental awareness & behavior                 | (Belan and Nenn, 2010; Conway et al., 2022; Meder and Kouma, 2010; Newburn and Alberini, 2016; Shin and McCann, 2018; Thurston et al., 2010; Ureta et al., 2021; Woodward et al., 2009) | –                                             | –                                                                                         |
|                   |                                                    | (Newburn and Alberini, 2016; Thurston et al., 2010; Ureta et al., 2021)                                                                                                                 | –                                             | –                                                                                         |
| Ability           | Financial incentive or income                      |                                                                                                                                                                                         | –                                             | –                                                                                         |
|                   | Cost                                               | –                                                                                                                                                                                       | –                                             | (Brown et al., 2016; Conway et al., 2022; Newburn et al., 2013; Shin and McCann, 2018)    |
|                   | Maintenance demand                                 | –                                                                                                                                                                                       | –                                             | Belan and Nenn (2010)                                                                     |
|                   | Spatial considerations                             | (Brown et al., 2016)                                                                                                                                                                    | –                                             | –                                                                                         |

(continued on next page)

**Table 1**

Authors' categorization of factors influencing household adoption of residential GSI.

| Motivational factors                                                         | Ability factors                                              |
|------------------------------------------------------------------------------|--------------------------------------------------------------|
| Sociodemographic attributes (age, educational level, self-identified gender) | Economic considerations (financial incentives, income, cost) |
| Social norms & aesthetics                                                    | Maintenance demand                                           |
| By-law enforcement & trust in municipal programs                             | Spatial considerations                                       |
| Direct benefits to household                                                 | Know-how                                                     |
| Environmental awareness & behavior                                           | Authority to make landscape changes                          |



Table 2 (continued)

| Type of influence | Independent variable                | Association from literature                                      |                             |                   |
|-------------------|-------------------------------------|------------------------------------------------------------------|-----------------------------|-------------------|
|                   |                                     | Positive                                                         | Non-Significant             | Negative          |
|                   | (space availability)                |                                                                  |                             |                   |
|                   | Know-how                            | (Brown et al., 2016; Conway et al., 2022; Woodward et al., 2009) | Newburn and Alberini (2016) | –                 |
|                   | Authority to make landscape changes | (Conway et al., 2022; Ureta et al., 2021)                        | –                           | Ren et al. (2020) |

other socio-demographic variables measured are residents' level of education and their stated gender. For a rain garden demonstration project, Belan and Nenn (2010) postulated that residents with university degrees were likely more informed about environmental issues, and consequently, this group had higher participation rates. But in another study in Maryland, USA, holding a college degree had no significant correlation with households' willingness to pay for a hypothetical rain garden rebate program (Newburn and Alberini, 2016). In regard to gender, two studies found that willingness to adopt a rain garden is positively correlated with survey respondents who selected female as their gender (Peng, 2010; Shin and McCann, 2018). A third study reported self-reported gender as a non-significant determinant, however,

those who self-identify as females were much more willing to install rain gardens than males if offered for free (Baptiste et al., 2015).

**3.3.1.2. Social norms and aesthetics.** Neighbours' acceptance of rain gardens is an extrinsic variable examined in three studies. The findings assert that neighbours of residents who adopted rain gardens expressed interest in rain garden programs and were more likely to adopt (Belan and Nenn, 2010; Shaw et al., 2011; Ureta et al., 2021).

Another extrinsic factor is the appearance of rain gardens. The results on the influence of aesthetics were inconclusive. During the operation and follow-up of a rain garden program in Wisconsin, USA, adopters and a number of their non-participating neighbours stated that they had expected "a more manicured look", and therefore some households replaced their native prairie garden with turf grass (Belan and Nenn, 2010, p.608). On the contrary, another case study, in Pennsylvania (USA), investigating the initial motivations of rain garden adopters concluded that the enhanced look of their property derived from a rain garden was selected as a motivating factor by 60% of the adopting households (Conway et al., 2022). Similarly, Newburn et al. (2013) found that rain garden appearance was not a barrier to adoption willingness, as only 5% of respondents did not like how rain gardens looked. For a hypothetical scenario of free rain garden installation, aesthetics and beautification did not affect the stated willingness to install rain gardens (Baptiste, 2014).

**3.3.1.3. By-law enforcement and trust in municipal services.** A study on a rain garden cost-share program compared the actual maintenance effort exerted by voluntary adopters and other households who were obliged

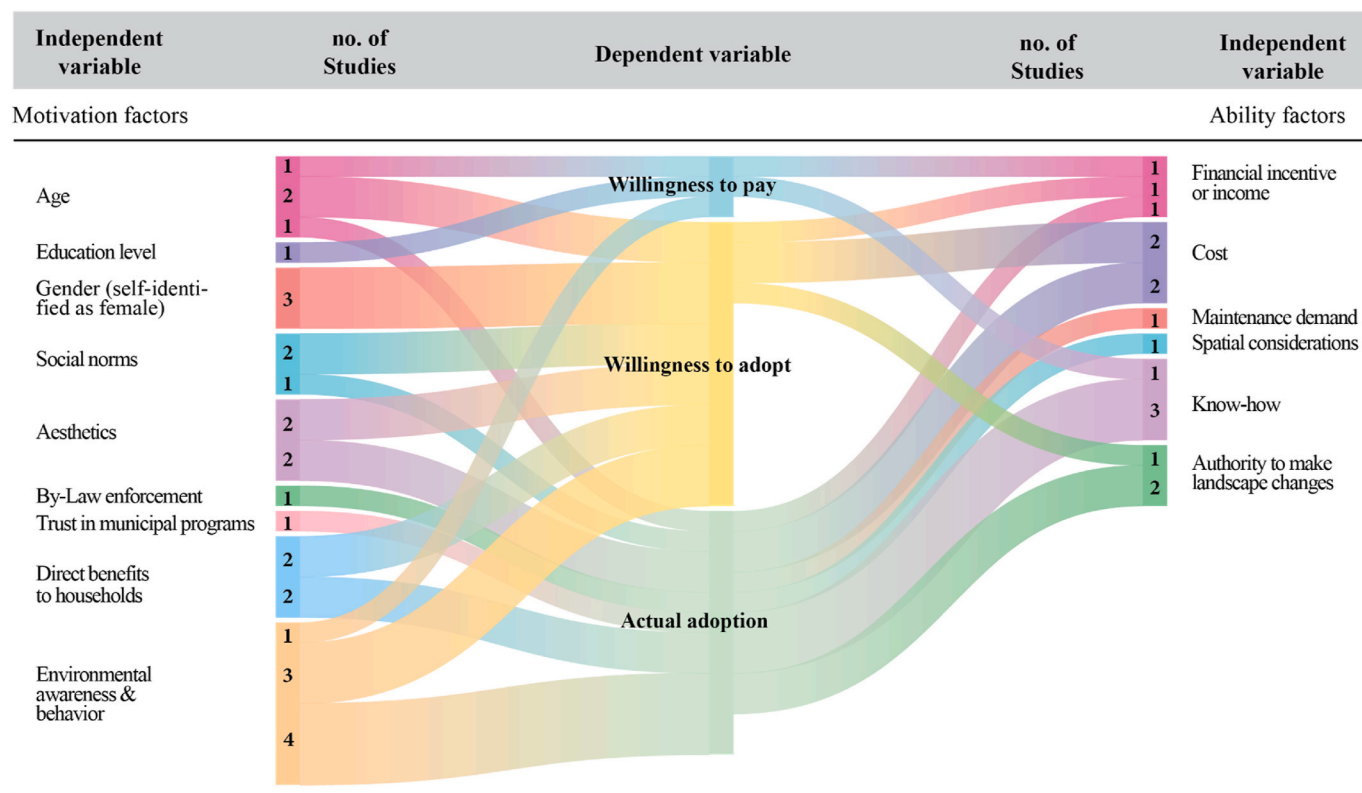


Fig. 2. Sankey Diagram summarizing the reviewed studies on rain gardens, the identified independent variables (left column: Motivational factors – right column Ability factors), the dependent variables, and the number of studies.

The thickness of the curves is relevant to the number of studies. For instance, under motivational factors (left column), the influence of "Environmental awareness & behavior" was addressed in three studies with relevance to "Willingness to adopt", in four studies with relevance to "Actual adoption", and in one study with relevance to "Willingness to pay". On the other hand, under "Ability factors" (right column) there is only one study that addressed "Maintenance demand" with relevance to "Actual adoption".

The total number of studies does not sum up to 100% because most studies examined more than one factor.

to install a rain garden (or other GSI practices) to meet the jurisdiction requirements of an impervious area limit (Woodward et al., 2009). Site inspections after rain garden installations revealed minimal maintenance efforts by the enforced group to avoid rain garden failure. Besides enforcement, the level of trust between residents and municipal government is another consideration for rain garden programs. For instance, a free provision rain garden program had low participation rates due to residents' distrust of city services (Belan and Nenn, 2010).

**3.3.1.4. Direct benefits to household.** We found mixed evidence about the influence of rain gardens to avoid flooding or rainwater pooling. Prior flooding experience had a consistent positive correlation with willingness to adopt (Ureta et al., 2021) and actual adoption of rain gardens (Ren et al., 2020). Similarly, a minority of households (33%) stated reducing storm flooding as a motivation to adopt rain gardens (Conway et al., 2022). However, Brown et al. (2016) reported a number of negative perceptions by residents of rain gardens contributing to water pooling and potentially damaging their properties and other properties nearby. In North Carolina, USA, many households were keen to participate in an initial workshop for a rain garden program. However, after learning that a rain garden alone would not resolve residents' flooding issue, they barely maintained their gardens, and the authors reported these installations among the failed ones (Woodward et al., 2009).

**3.3.1.5. Environmental awareness and behavior.** Environmental attitude and concern about local ecological assets (Conway et al., 2022; Shin and McCann, 2018; Ureta et al., 2021; Woodward et al., 2009), as well as environmental awareness because of outreach and community educational programs, were found to be positive predictors for actual rain garden implementation or favorable intentions (Belan and Nenn, 2010; Thurston et al., 2010). Similarly, Newburn and Alberini (2016) found that respondents with high levels of environmental concern about their local water streams were willing to pay more for a hypothetical rebate program in Maryland, USA. More specifically, we found evidence that pro-environmental behaviors, such as the prior adoption of GSI, are strong predictors of rain garden adoption. Further, Meder and Kouma (2010) surveyed participants of a rain garden program in South Central Wisconsin, USA, and concluded that rain garden adopters expressed interest in installing an additional GSI on their property.

### 3.3.2. Ability factors for rain garden adoption

**3.3.2.1. Financial incentive or income, cost.** The available evidence demonstrates that financial incentives can facilitate rain garden adoption or increase households' willingness to accept its implementation on their properties (Thurston et al., 2010). Further, relatively higher-income households were more likely to install a rain garden (Ureta et al., 2021). Income was also positively and significantly associated with the tendency to pay more per square foot to implement a rain garden (Newburn and Alberini, 2016).

We observed a relevant trend with respect to the expenses incurred from installation and maintenance. When examining cost influence on adoption likelihood, households' responses in three different studies indicated high cost as a barrier to consider installing a rain garden (Conway et al., 2022; Newburn et al., 2013; Shin and McCann, 2018), especially when considering the tradeoff between personal direct benefits and the high cost of implementation (Brown et al., 2016). A negative correlation was also reported between residents' adoption willingness and the expected size of a rain garden; the bigger the rain garden, the higher the implementation cost, which consequently lowered a household's adoption willingness (Newburn et al., 2013).

**3.3.2.2. Spatial considerations, and maintenance demand.** Space availability can also hinder household adoptions. For example, Brown et al.

(2016) found that the loss of garden spaces was a common negative perception of rain gardens. For this factor, further clarification is required to determine whether space availability is a limiting (i.e., ability) factor or a motivational one. Yard size can arguably be a motivational factor in case residents have enough space for a rain garden but prefer to conserve their yard space for other uses. Regardless of yard size, rain garden size can also influence residents' perceptions. For example, Belan and Nenn (2010) noted that some rain garden adopters sought to shrink their rain garden to limit the required maintenance effort.

**3.3.2.3. Know-how.** The role of technical knowledge was also explicitly investigated. Participants in a stormwater retrofit program stated prior knowledge of infiltration processes as a primary motivation to opt for rain garden installation (Brown et al., 2016). Furthermore, some interviewees credited their installation willingness to their occupational or educational background while others noted their lack of information on the function and operationalization of rain gardens as a barrier to installing them on their yards. Newburn and Alberini (2016) concluded that prior knowledge of rain gardens as a stormwater management practice was an insignificant indicator of the willingness to pay for rain garden installation.

Prior gardening experience is another consideration that relates to residents' technical knowledge level and incurred costs. Newburn et al. (2013) suggested that households with gardening experience could have enough knowledge to install and maintain their gardens without the need to hire a landscape contractor and therefore the overall cost would be lower. According to Woodward et al. (2009), rain gardens owned by active "gardeners" seemed to function well and generally had high success rates. Conway et al. (2022) findings support that gardening knowledge decreased the perceived risk associated with maintenance requirements, consequently increasing rain garden uptake.

**3.3.2.4. Authority to make landscape changes.** Some scholars used homeownership as a proxy for household decision-making authority; hence, we consider it a factor that could limit a household's ability to have a rain garden on their yards. In this context, "owner" refers to households who own and live on the property. When examining statistical significance, we found inconsistent results. One study found that renters were more likely (than owners) to be aware of flooding issues in their neighbourhoods and keener to install rain gardens (Ren et al., 2020). The authors attributed these results to areas where rental properties are possibly more prone to flooding issues, and hence, renters were interested in installing rain gardens to address their runoff concerns. On the contrary, Ureta et al. (2021) concluded that respondents with the authority to manage their landscapes (i.e., owners) were more likely to adopt rain gardens. In Philadelphia, USA, none of the renters who participated in a survey for a GSI program had a rain garden (Conway et al., 2022).

### 3.4. Adoption determinants for rainwater barrels or tanks

Similar to rain gardens, we identified 14 studies that investigate predictors for rainwater tank or barrel adoptions. Slightly more than half (57%) of the studies measured the likelihood of residents installing this practice ( $n = 8$ ) and others analyzed actual adoption as a dependent variable ( $n = 6$ ) (Table 3, Fig. 3).

#### 3.4.1. Motivational factors for rainwater tanks and barrel adoption

**3.4.1.1. Sociodemographic attributes.** We identified three demographic characteristics used as explanatory factors for rain barrel adoptions. Two studies reported that age had a significant negative association with households' willingness to install a rain barrel in the USA (Shin and McCann, 2018; Ureta et al., 2021). In contrast, one other study reported

**Table 3**

Adoption determinants for rainwater tanks or barrels and their associations with actual adoption, or willingness to adopt.

| Type of influence | Independent variable                                     | Associations from literature                                                                                                                                                          |                                       |                                             |
|-------------------|----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|---------------------------------------------|
|                   |                                                          | Positive                                                                                                                                                                              | Non-Significant                       | Negative                                    |
| Motivation        | Age                                                      | Ren et al. (2020)                                                                                                                                                                     | Du Toit and Chilwane (2022)           | (Shin and McCann, 2018; Ureta et al., 2021) |
|                   | Education level                                          | Gao et al. (2016)                                                                                                                                                                     | Du Toit and Chilwane (2022)           | –                                           |
|                   | Gender (self-identified as female)                       | (Gao et al., 2016; Shin and McCann, 2018)                                                                                                                                             | –                                     | –                                           |
|                   | Social norms                                             | Ureta et al. (2021)                                                                                                                                                                   | –                                     | –                                           |
|                   | Aesthetics                                               | –                                                                                                                                                                                     | Meenar et al. (2022)                  | (Brown et al., 2016; Conway et al., 2022)   |
|                   | By-Law enforcement & Trust in municipal programs         | (Brown et al., 2016)                                                                                                                                                                  | –                                     | –                                           |
|                   | Direct Household Benefits                                | (Bakacs et al., 2013; Brown et al., 2016; Gao et al., 2016; Newburn et al., 2013)                                                                                                     | (Baptiste, 2014; Conway et al., 2022) | –                                           |
|                   | Environmental awareness & behavior                       | (Ando and Freitas, 2011; Bakacs et al., 2013; Conway et al., 2022; Du Toit and Chilwane, 2022; Gao et al., 2016; Meder and Kouma, 2010; Shin and McCann, 2018; Thurston et al., 2010) | –                                     | –                                           |
|                   | Financial incentive or income                            | (Ando and Freitas, 2011; Brown et al., 2016)                                                                                                                                          | Thurston et al. (2010)                | –                                           |
|                   | Cost                                                     | –                                                                                                                                                                                     | Gao et al. (2016)                     | Shin and McCann (2018)                      |
| Ability           | Maintenance demand                                       | –                                                                                                                                                                                     | Baptiste (2014)                       | (Gao et al., 2016; Meenar et al., 2022)     |
|                   | Spatial considerations (space availability)              | (Brown et al., 2016; Meenar et al., 2022; Ren et al., 2020)                                                                                                                           | –                                     | –                                           |
|                   | Spatial considerations (distance to distribution centre) | –                                                                                                                                                                                     | –                                     | Ando and Freitas (2011)                     |
|                   | Know-how                                                 | (Brown et al., 2016; Gao et al., 2016)                                                                                                                                                | –                                     | –                                           |

**Table 3 (continued)**

| Type of influence | Independent variable                | Associations from literature                                 |                 |          |
|-------------------|-------------------------------------|--------------------------------------------------------------|-----------------|----------|
|                   |                                     | Positive                                                     | Non-Significant | Negative |
|                   | Authority to make landscape changes | (Ando and Freitas, 2011; Gao et al., 2016; Ren et al., 2020) | –               | –        |

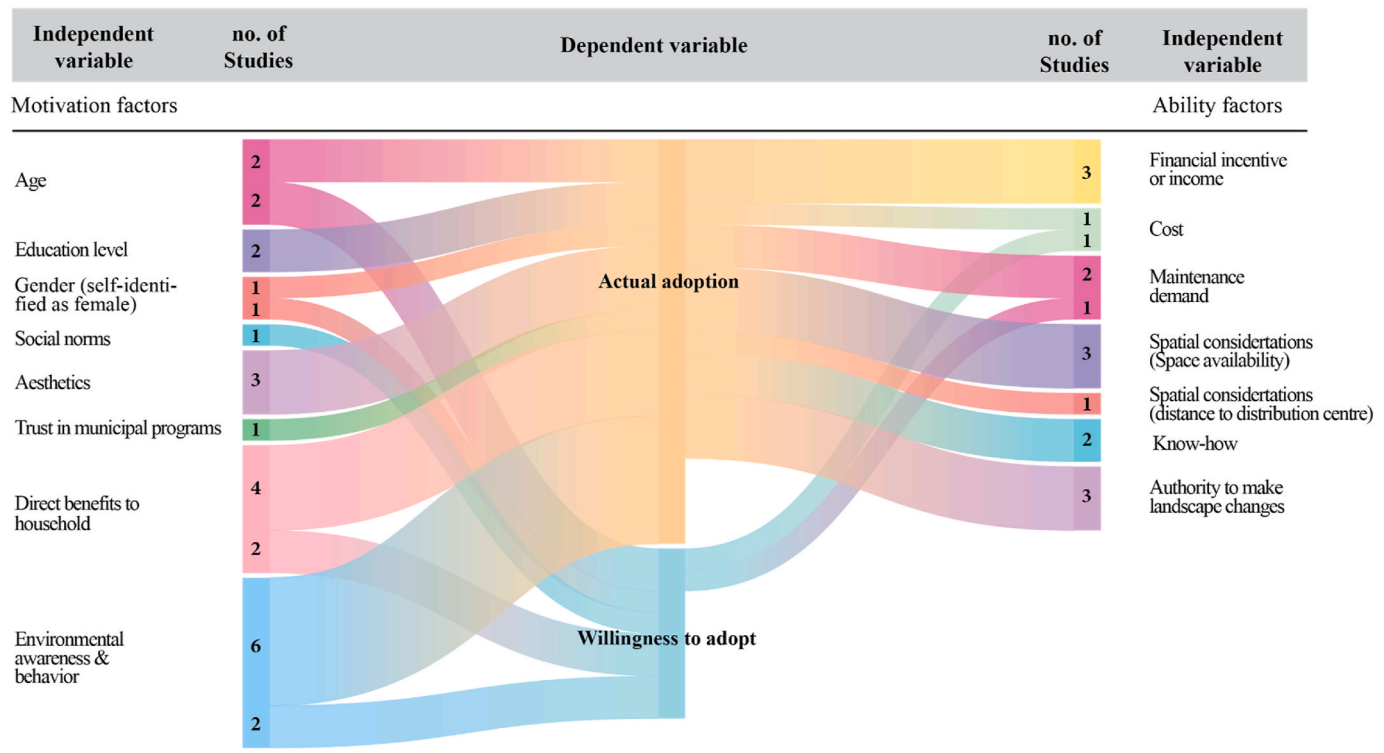
age as an insignificant predictor for the adoption of rainwater tanks in South Africa (Du Toit and Chilwane, 2022). A study by Ren et al. (2020) even suggests a positive correlation, as residents who were 58 years or older were more likely than younger residents to adopt roof diversion to a rain barrel (Ren et al., 2020). Second, the level of residents' education varied from positively correlated with rain barrel adoption (Gao et al., 2016) to insignificant for rainwater harvesting tanks (Du Toit and Chilwane, 2022). The third demographic attribute influencing the likelihood of rain barrel adoption was the resident's self-identified gender. Respondents identifying themselves as female were positively associated with rain barrel installations (Gao et al., 2016), and self-identified males were less likely to be potential adopters (Shin and McCann, 2018).

**3.4.1.2. Social norms and aesthetics.** The presence of neighbours who are adopting rain barrels is a positive predictor for adoption of this GSI practice. Empirical evidence demonstrates that households near rain barrel adopters are relatively more willing to adopt a rain barrel as well (Ureta et al., 2021).

Three studies explicitly investigated whether and how appearance influenced households' decisions to adopt rain barrels on their yards. Meenar et al. (2022) found contrasting perceptions of rain barrels. Rain barrel adopters did not report negative perceptions about the look of rain barrels. However, free provision and creative painting workshops were not enough for non-adopters to change their mind, who stated their reluctance to adopt them because of their unpleasant look, among other factors. The findings of Conway et al. (2022) support households' negative perceptions of rain barrels, especially when compared to trees and rain gardens. Additionally, Brown et al. (2016) noted residents' negative perceptions of the look of rainwater tanks and their associated connections, which one resident identified as a potential reason for reducing their property value. Above-ground rainwater tanks are hard to blend into a yard landscape due to their large size.

**3.4.1.3. Direct benefits to household.** Besides their environmental contribution in controlling runoff at source, our findings suggest direct benefits related to households' daily needs as a positive motivational factor for rain barrel adoption. For example, Bakacs et al. (2013) conducted follow-up surveys to study a four-year program in Northern Virginia and New Jersey, USA. Residents stated that "saving water" was their primary motivation for acquiring rain barrels. Practicing flower and vegetable gardening also was a positive, strong predictor of rain barrel adoption likelihood (Newburn et al., 2013), which suggests that collecting rainwater for watering gardens is a motivational factor for rain barrel adoption. Two more studies provided evidence that watering vegetable or flower gardens is a positive motivational factor for actual adoptions. Brown et al. (2016) found several household direct benefits of rain barrels, but using collected rainwater for watering flower and vegetable gardens was the top motivation. Similarly, in Indiana, USA, over 90% of the survey respondents in two different watersheds said they harvested rainwater to irrigate their gardens (Gao et al., 2016).

A minority of rain barrel adopters stated mitigating rainwater pooling or flooding damage to their properties as a motivating adoption factor (Conway et al., 2022). In contrast, Baptiste (2014) found that collecting rainwater for gardening purposes and protecting house



**Fig. 3.** Sankey Diagram summarizing the reviewed studies on rainwater tanks/barrels, the identified independent variables (left column: Motivational factors – right column Ability factors), the dependent variables, and the number of studies.

The thickness of the curves is relevant to the number of studies. For instance, under motivational factors (left column), the influence of “Trust in municipal programs” was addressed in one study in relevance with “Actual adoption”. Similarly, under “Ability factors” (right column) two studies addressed “Know-how” in relevance with Aactual adoption”.

The total number of studies does not sum up to 100% because most studies examined more than one factor.

foundations was not significantly correlated with households’ willingness to adopt this GSI practice.

**3.4.1.4. Environmental awareness and behavior.** Outreach programs and demonstration projects positively influence rain barrel adoption in different contexts. For a GSI program in Ohio, USA, study results suggested that an education campaign run before the program caused a relatively high number of bidders (around 55%) who opted to participate in an auction without requesting any subsidy (Thurston et al., 2010). Another stormwater educational program run in Chicago, USA, targeted households living in proximity to Chicago’s Green Alleys, where permeable pavers incorporate stormwater retention trenches (Ando and Freitas, 2011). The outreach information had a positive yet weak influence on the rates of rain barrel adoption. The authors suggested that GSI installations on nearby public land might have minimized households’ sense of responsibility and, consequently, their motivation to install rain barrels on their private properties (Ando and Freitas, 2011).

Several studies suggested pro-environmental behaviors and attitudes as positive predictors of rain barrel adoption. Such behaviors included “green party” voting in the USA (Ando and Freitas, 2011), water-saving practices such as dual flush toilets in South Africa (Du Toit and Chilwane, 2022), high levels of trust in environmental organizations as a proxy for environmental attitudes (Shin and McCann, 2018), and prior rain barrel adoption in the USA (Gao et al., 2016). Meder and Kouma (2010) found an association between rain barrel awareness and uptake in a free rain barrel provision program. The authors documented an increasing percentage of residents aware of rain barrels from 16% to 24% in one year, accompanied by a significant rise in rain barrel sales. These results are consistent with other evidence asserting the positive associations between households’ knowledge of the watershed concept

and their willingness to adopt rain barrels (Shin and McCann, 2018), and a similar influence between residents concerned about environmental issues and adopting a rain barrel (Conway et al., 2022; Gao et al., 2016). Studies evaluating previous GSI programs suggested that rain barrel adopters were willing to adopt an additional rain barrel or a rain garden (Bakacs et al., 2013; Meder and Kouma, 2010).

#### 3.4.2. Ability factors for rainwater tanks and barrel adoption

**3.4.2.1. Financial incentives, income and cost.** Ando and Freitas (2011) investigated spatial patterns of adoption related to a subsidized rain barrel program in Chicago, USA, and observed more purchases within higher-income neighbourhoods. Adopters of rainwater tanks also identified financial incentives as the salient uptake factor for a GSI program in Melbourne, Australia (Brown et al., 2016). This finding was underpinned by the fact that rainwater tanks in Melbourne were relatively expensive at the time. Besides cost subsidies, households identified savings on water bills and mitigating fluctuating water costs as complementary financial benefits. The results from the Melbourne case study may seem to contradict the findings from an auction for a GSI program in Ohio, USA, where most residents did not request a subsidy to install a rain barrel (Thurston et al., 2010). However, compared to rain barrels in Ohio (USA \$ 130 in 2010), rainwater tanks in the Melbourne case were significantly more costly (around AUS \$ 6000–7000 in 2016, including installation) because they had larger storage capacity and required professional plumbing connections. Longer, drier periods and higher temperatures in Melbourne were additional contextual differences, which may explain why residents were more concerned about minimizing domestic water costs.

We observed mixed results regarding the influence of rain barrel cost on adopter and non-adopter residents. On the one hand, Gao et al.



(2016) asserted that less than one quarter of non-adopter respondents (20%) identified cost as a barrier to rain barrel adoption, while the majority of adopters (>90%) stated that they installed rain barrels because they consider purchasing and maintaining rain barrels to be financially feasible. On the other hand, [Shin and McCann \(2018\)](#) found that cost is a potential barrier for both persistent non-adopters and likely adopters, even though the latter group was more concerned about it. In this case study, factors beyond cost led to non-adopters decisions; lower costs, it seemed, would not overcome these other factors.

**3.4.2.2. Maintenance demand.** [Meenar et al. \(2022\)](#) reported non-adopter concerns about the time required to maintain rain barrels and [Gao et al. \(2016\)](#) found that technical malfunction was the top constraint to the continued use of rain barrels. However, another case study found that maintenance was an insignificant predictor of adoption willingness ([Baptiste, 2014](#)).

**3.4.2.3. Spatial considerations.** Several studies asserted that the size of residents' yards or parcels can limit rain barrel adoptions. The findings indicate that households were unwilling to trade off their leisure space for rainwater tank installation, which are usually large structures and occupy a considerable area ([Brown et al., 2016](#)). Although rain barrels are relatively smaller than rainwater tanks, residents from New Jersey, USA, questioned the viability of installing rain barrels in limited-area private open spaces ([Meenar et al., 2022](#)). Similarly, [Ren et al. \(2020\)](#) found that rain barrel adoptions were positively correlated with larger residential parcels. Referring to ease of access, [Ando and Freitas \(2011\)](#) found that census tracts in proximity to a rain barrel distribution centre had more rain barrel adopters. The authors suggested that facilitating the delivery of rain barrels to residents' properties could significantly increase the number of adopters. The authors also speculated that other factors may have complemented the influence of the location of rain barrel distribution centres. For instance, some centres were part of broader advertising efforts for rain barrel distributions, such as news articles and information about rain barrel benefits attached to water bills.

**3.4.2.4. Know-how.** Technical knowledge related to planting crops or rainwater harvesting facilitates the adoption of rain barrels. For instance, in a study in Victoria, Australia, four out of 15 interviewees (26%) accredited their familiarity with rainwater collection practices to their farming background ([Brown et al., 2016](#)). Other adopters identified the lack of operational information for rain barrels, such as how to deal with water pressure issues, as one of the top three constraints to maintaining a rain barrel ([Gao et al., 2016](#)). Further, non-adopters were reluctant to install rain barrels on their yards because they did not have the knowledge or the time to install and maintain them.

**3.4.2.5. Authority to make landscape changes.** Three studies consistently reported a positive association between homeownership and the likelihood or actual adoption of rain barrels. [Ando and Freitas \(2011\)](#) found a positive correlation between owner occupancy and rain barrel purchases in Chicago, USA. The authors further suggested that occupants of multiunit buildings are less likely to install rain barrels because of limited indoor use of rainwater. Similarly, actual adoption was more likely for survey respondents who owned their property than renters ([Ren et al., 2020](#)). Owners were also more likely to adopt rain barrels in another study ([Gao et al., 2016](#)).

### 3.5. Adoption determinants for permeable pavers

Few studies addressed the adoption of permeable pavers ( $n = 5$ ) compared to rain gardens and rainwater tanks and barrels. Two studies (40%) investigated actual adoption, and two studies (40%) addressed residents' stated willingness to adopt permeable pavers. Only one study

examined the willingness to pay for installing permeable pavers ([Table 4](#), [Fig. 4](#)).

#### 3.5.1. Motivational factors for permeable pavers adoption

**3.5.1.1. Social norms.** Only one socio-psychological barrier was addressed in relevance to permeable paver uptake: social acceptability. For a study in Ontario, Canada, [Cote and Wolfe \(2014\)](#) inferred that social norms could have hindered the diffusion of permeable pavers adoption. Their inference was based on survey results indicating that although the income of most respondents (90%) ranged between high to moderate, the majority stated they "might" rather than "would" consider installing permeable pavers. However, the authors acknowledged the possibility that other barriers might have exacerbated such social reluctance, such as residents' limited awareness of this practice and perceived risks of basement flooding or groundwater contamination ([Cote and Wolfe, 2014](#)).

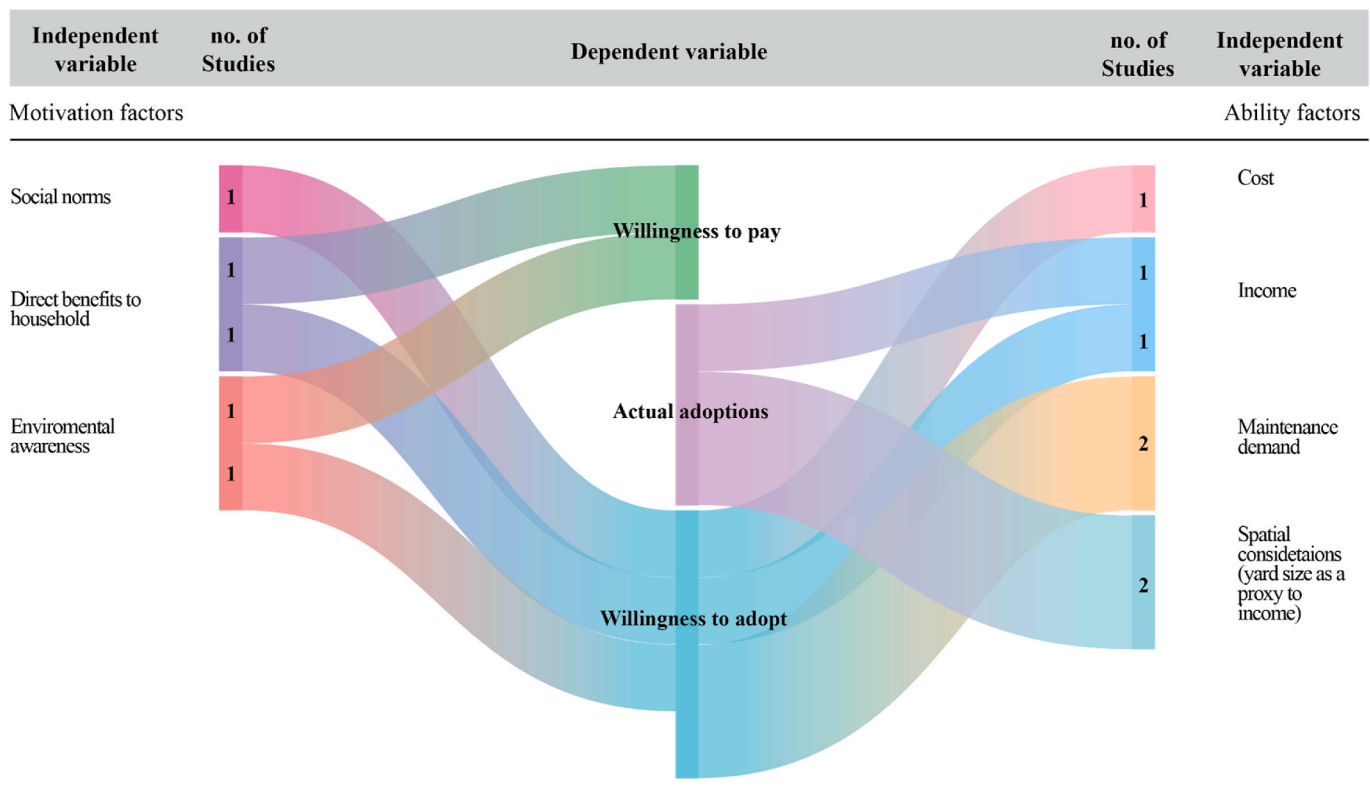
**3.5.1.2. Direct benefits to household.** Other motivations for permeable pavers adoption included direct household benefits such as controlling pooling water and ice buildup; however, these factors were not statistically significant ([Baptiste, 2014](#)). Interestingly, the evidence indicated that households with flower gardens reported a higher willingness to pay for installing permeable pavers than other infiltration practices such as a dry well or infiltration trenches in Pennsylvania, USA ([Fleming and Savchenko, 2022](#)). The authors of this study further suggested that the influence of explanatory variables, such as owning a flower garden, might vary according to a households' perceived benefits of the different GSI practices ([Fleming and Savchenko, 2022](#)). Perhaps, in this case, flower gardeners prioritize the aesthetic dimension of permeable pavers, especially if the paved area is used as a patio. Hence, they are more likely to consider paying for permeable pavers rather than underground interventions such as dry wells or infiltration trenches. Further, respondents who stated permeable pavers as their preferred GSI practice (i.e., more preferred than rain gardens and dry wells) expressed concerns about stormwater runoff as the most significant explanatory factor for the willingness to pay ([Fleming and Savchenko, 2022](#)).

**3.5.1.3. Environmental awareness.** Our findings indicate that environmental knowledge or awareness of flooding issues are influential

**Table 4**

Adoption determinants for permeable pavers and their associations with actual adoption, willingness to adopt, or pay.

| Type of influence | Independent variable                                    | Association from literature                                                   |                                       |                                       |
|-------------------|---------------------------------------------------------|-------------------------------------------------------------------------------|---------------------------------------|---------------------------------------|
|                   |                                                         | Positive                                                                      | Non-Significant                       | Negative                              |
| Motivation        | Social norms                                            | <a href="#">Cote and Wolfe (2014)</a>                                         | –                                     | –                                     |
|                   | Direct Household benefits                               | <a href="#">Fleming and Savchenko (2022)</a>                                  | <a href="#">Baptiste (2014)</a>       | –                                     |
|                   | Environmental awareness                                 | <a href="#">Cote and Wolfe, 2014</a> ; <a href="#">Ren et al., 2020</a>       | –                                     | –                                     |
| Ability           | Cost                                                    | –                                                                             | –                                     | <a href="#">Cote and Wolfe (2014)</a> |
|                   | Income                                                  | <a href="#">Ren et al. (2020)</a>                                             | –                                     | –                                     |
|                   | Maintenance demand                                      | –                                                                             | <a href="#">Cote and Wolfe (2014)</a> | <a href="#">Baptiste (2014)</a>       |
|                   | Spatial considerations (yard size as a proxy to income) | <a href="#">Du Toit and Chilwane, 2022</a> ; <a href="#">Ren et al., 2020</a> | –                                     | –                                     |



**Fig. 4.** Sankey Diagram summarizing the reviewed studies on permeable paving, the identified independent variables (left column: Motivational factors – right column Ability factors), the dependent variables, and the number of studies.

The thickness of the curves is relevant to the number of studies. For instance, under motivational factors (left column), the influence of “Social norms” was addressed in one study in relevance with “Willingness to adopt”. Similarly, under “Ability factors” (right column), one study addressed “Cost” in relevance with “Willingness to adopt”.

The total number of studies does not sum up to 100% because most studies examined more than one factor.

motivational factors in residents’ willingness to implement permeable pavers. For example, [Cote and Wolfe \(2014\)](#) found that most survey respondents concerned about improving stormwater management efficiency within their neighbourhood were more likely to consider implementing permeable pavers. Further, results from another survey examining factors for actual adoption suggest that permeable paver adoption is associated with the recognition of runoff issues ([Ren et al., 2020](#)).

### 3.5.2. Ability factors for rain barrel adoption

**3.5.2.1. Cost, income, and spatial considerations.** Although permeable pavers do not limit open space use, we found a positive correlation between the size of residential properties and permeable pavers adoption ([Du Toit and Chilwane, 2022](#); [Ren et al., 2020](#)). In these cases, it may be that property size is an indicator of higher income, which is positively correlated with permeable paver adoption ([Ren et al., 2020](#)). In addition, [Cote and Wolfe \(2014\)](#) found that none of their survey respondents were willing to pay more than 50% on top of what conventional driveways would cost, and they would only consider spending more if the city ran cost-share programs.

**3.5.2.2. Maintenance demand.** Maintenance is another factor that was reported as potentially limiting residents’ ability to adopt permeable pavers. However, our findings were inconsistent in this regard. On the one hand, [Baptiste \(2014\)](#) asserted that maintenance is a statistically significant negative determinant of residents’ willingness to install permeable pavers. On the other hand, most respondents in a study by [Cote and Wolfe \(2014\)](#) stated a positive willingness to perform annual

maintenance of their permeable pavers, leading the authors to suggest that maintenance is not an impeding factor. The authors supported their conclusion with literature evidence stating that annual maintenance for some permeable paver types (such as concrete lattice) might not be required ([Brattebo and Booth, 2003](#)).

## 4. Discussion

Our findings include key theoretical recommendations for GSI scholars and practical implications for residential GSI policymaking and practice to achieve higher household uptake. Guided by the Fogg behavioral model constructs, we structure our discussion in three stages. First, we reflect on motivational and ability factors, then synthesize their interrelationships, and finally, discuss policy tools as adoption triggers.

### 4.1. Reflections on differences and commonalities among GSI types

Our comparative analysis results support our hypothesis that adoption factors could vary in direction and significance from one GSI practice to another. Beyond our initial supposition, a specific determinant can have inconsistent effects on the same dependent variable for the same practice in different contexts. An example of such varying influences of motivational factors is aesthetics, which had inconsistent associations with the actual adoption of rain gardens and rainwater tanks or barrels. Inconsistencies also persist among the different dependent variables for the same GSI practice. For example, our findings suggest that rain garden aesthetics was an insignificant predictor for household adoption willingness, while it was reported as a positive motivation for actual adopters in a different case study, and as a cause for adoption termination in another study. We also identified similar

variations for ability factors. For instance, cost influence varied from insignificant to negative association, depending on the size and type of the GSI intervention.

These variations underscore the critical need for specificity in identifying the type of GSI practice, the dependent variable investigated, and the corresponding explanatory factor. From a socio-psychological perspective, Ajzen (2020) emphasized the principle of "compatibility," which implies that behavioral predictors should precisely reflect the involved "action," "target," or purpose, spatial or social "context," and the "time frame." Parallel to this, studies investigating households' private-yard decisions suggest that correlations between environmental values and pro-environmental behaviors are not universal. For example, environmental values can sometimes be correlated with behaviors that could degrade the environment, such as the use of pesticides and improper irrigation patterns (Robbins and Sharp, 2003; Turner et al., 2016; Yabiku et al., 2008). Therefore, using relevant yet not specific pro-environmental behavioral predictors, such as practicing rainwater harvesting, as a proxy to predict general GSI adoption is potentially an improper extrapolation. To this end, we call on scholars to avoid using findings from one GSI practice to reflect on GSI generally in their discussions or conclusions. Similarly, policymakers, environmental non-profits, and involved professionals should be mindful of such subtle yet impactful differences when designing or implementing GSI policies, programs, or interventions on private residential yards.

The comparison also revealed some commonalities in motivational and ability factors. One example, under the motivation category, is the positive association between peer influence or social acceptability and residents' willingness to adopt rain gardens, rainwater tanks or barrels, and permeable pavers. This finding implies that achieving a critical mass of adopters is essential to avoid a failed diffusion of the three practices (Rogers, 2003). Homeownership was also reported as an enabling factor for households' willingness or actual adoption of rain gardens and rainwater tanks or barrels. Together, these two findings align with the rationale of GSI program managers targeting neighbourhoods with higher owner-occupancy rates (Belan and Nenn, 2010) and scholars studying residential areas with a majority of single-family dwellings (Newburn and Alberini, 2016). However, efforts should be made to address equity concerns and to avoid marginalization of non-homeowner residents. Additionally, programs should promote the use of informational signage on adopters' yards to induce further adoption by neighbours (Gao et al., 2016). Recognition and award programs can incentivize such promotional efforts.

Furthermore, we found more consensus on the significance of predictors that primarily relate to egoistic, hedonic values than altruistic and biospheric determinants. This finding aligns with previous research on the influence of personal environmental values on household yard management decisions, such as vegetation choices and fertilization

(Larson et al., 2010). In the context of residential GSI, examples of such an anthropocentric behavioral approach include prioritizing direct benefits such as saving costs over protecting nearby water bodies for rainwater barrels or tanks (Bakacs et al., 2013) and pride of property in relevance to aesthetics for both rain gardens and rainwater barrels or tanks (Belan and Nenn, 2010; Brown et al., 2016). To this end, we suggest that public outreach programs should address positive environmental impacts and, perhaps more importantly, GSI's direct benefits with a focus on potential financial savings or increased property values.

Finally, when comparing motivational and ability factors (Table 5 and 6), it is evident that the number of explanatory factors under "motivations" and the number of studies addressing them (refer to Figure 2, 3 and 4) is more than those under "abilities," which may indicate that more literature has addressed household motivations. Further reflecting on Table 5 and 6, we note that social norms and environmental attitudes/behaviors are positive predictors across the three GSI practices. However, we found mixed results regarding the influence of cost and maintenance under the abilities group. This finding aligns very closely with another study that applied the Theory of Planned Behavior to guide the design of stormwater management outreach programs. Shaw et al. (2011) found that both attitudes and norms were positively associated with their survey respondents' intentions to install a rain garden. Yet, similar to our findings, the third construct of the TPB, perceived behavioral control, was not associated. However, the authors also stated that the influence of behavioral control – representing household perceived ability – may vary according to the context and type of intention under investigation (Shaw et al., 2011). In this context, further investigation could consider whether the actual adoption of GSI might be affected by perceived ability while the intent might not be affected, or vice versa (see Table 6).

#### 4.2. GSI adoption results from converging ability and motivational factors

Although the comparative analysis provides important insights into inconsistencies and common trends in associations between the different dependent and explanatory adoption factors, we argue that looking at predictors in isolation oversimplifies household adoption dynamics and potentially limits the practicality of the findings. To this end, we use the Fogg model to qualitatively illustrate how conflicting and complementary factors may lead to residential GSI rejection or adoption.

Some programs designed to increase GSI adoption integrated educational campaigns to provide households with tools that address their subjective motivations about the benefits of installing GSI on private yards and provide economic incentives to address their financial risks. Although such programs are sometimes relatively successful, potential adopters stated concerns that hindered their adoption. For instance, most respondents (69%) for a study in Melbourne, Australia,

**Table 5**

Evidence for the influence of motivational factors: Differences and commonalities between GSI practices.

| Independent variable                             | Dependent variables |     |     |                            |      |                  |     |     |                        | Inferred influence on the three practices |
|--------------------------------------------------|---------------------|-----|-----|----------------------------|------|------------------|-----|-----|------------------------|-------------------------------------------|
|                                                  | Rain gardens        |     |     | Rainwater tanks or barrels |      | Permeable Pavers |     |     |                        |                                           |
|                                                  | A                   | W   | WTP | A                          | W    | A                | W   | WTP |                        |                                           |
| Age                                              | –                   | –   | –   | Inc.                       | –    | NA               | NA  | NA  | More evidence required |                                           |
| Educational level                                | NA                  | NA  | N.S | Inc.                       | NA   | NA               | NA  | NA  | More evidence required |                                           |
| Gender (self-identified as female)               | +                   | N.S | NA  | +                          | +    | NA               | NA  | NA  | More evidence required |                                           |
| Social norms                                     | +                   | +   | NA  | NA                         | +    | NA               | +   | NA  | +                      |                                           |
| Aesthetics                                       | Inc.                | N.S | NA  | Inc.                       | NA   | NA               | NA  | NA  | More evidence required |                                           |
| By-Law Enforcement & Trust in municipal programs | +                   | NA  | NA  | +                          | NA   | NA               | NA  | NA  | More evidence required |                                           |
| Direct benefits to household                     | Inc.                | +   | NA  | Inc.                       | Inc. | NA               | N.S | +   | Inconsistent           |                                           |
| Env. Awareness & Behavior                        | +                   | +   | +   | +                          | +    | +                | +   | NA  | +                      |                                           |

– Dependent variables are as follows: A: Actual Adoption, W: Willingness to adopt, WTP: Willingness to pay.

– Associations: (+): positive, (–): Negative, Inc.: Inconsistent, N.S: Non-significant, NA: Not applicable.

– We infer the cumulative association for each independent variable as positive or negative when the association direction is consistent for one or more dependent variable(s) across the three practices. Additionally, we do not account for non-significant associations.

– "More evidence required" indicates missing evidence for one or more GSI practice(s).

**Table 6**

Evidence for the influence of ability factors: Differences and commonalities between different GSI adoptions.

| Independent variable                | Dependent variables |     |     |                         |     |  |                  |      |     | Inferred influence on the three practices |
|-------------------------------------|---------------------|-----|-----|-------------------------|-----|--|------------------|------|-----|-------------------------------------------|
|                                     | Rain gardens        |     |     | Rainwater tanks/barrels |     |  | Permeable Pavers |      |     |                                           |
|                                     | A                   | W   | WTP | A                       | W   |  | A                | W    | WTP |                                           |
| Financial incentive/income          | +                   | +   | +   | Inc.                    | NA  |  | +                | +    | NA  | +                                         |
| Cost                                | –                   | –   | NA  | N.S                     | –   |  | NA               | –    | NA  | Inconsistent                              |
| Maintenance demand                  | –                   | N.S | NA  | –                       | N.S |  | NA               | Inc. | NA  | Inconsistent                              |
| Spatial considerations              | +                   | NA  | NA  | +                       | NA  |  | +                | NA   | NA  | +                                         |
| Know-how                            | +                   | NA  | N.S | +                       | NA  |  | NA               | NA   | NA  | More evidence required                    |
| Authority to make landscape changes | Inc.                | +   | NA  | +                       | NA  |  | NA               | NA   | NA  | More evidence required                    |

– Dependent variables are as follows: A: Actual Adoption, W: Willingness to adopt, WTP: Willingness to pay.

– Associations: (+): positive, (–): Negative, Inc.: Inconsistent, N.S: Non-significant, NA: Not applicable.

– We infer the cumulative association for each independent variable as positive or negative when the association direction is consistent for one or more dependent variable(s) across the three practices. Additionally, we do not account for non-significant associations.

– “More evidence required” indicates missing evidence for one or more GSI practice(s).

stated their independence from outdoor water use restrictions, such as watering their gardens, as their primary perceived personal benefit of installing rainwater tanks. A cost-share program further complemented this positive perception. However, 14 out of 35 residents considering participating did not install the tanks due to a lack of trust in the program (Brown et al., 2016) (Fig. 5, Case Study A).

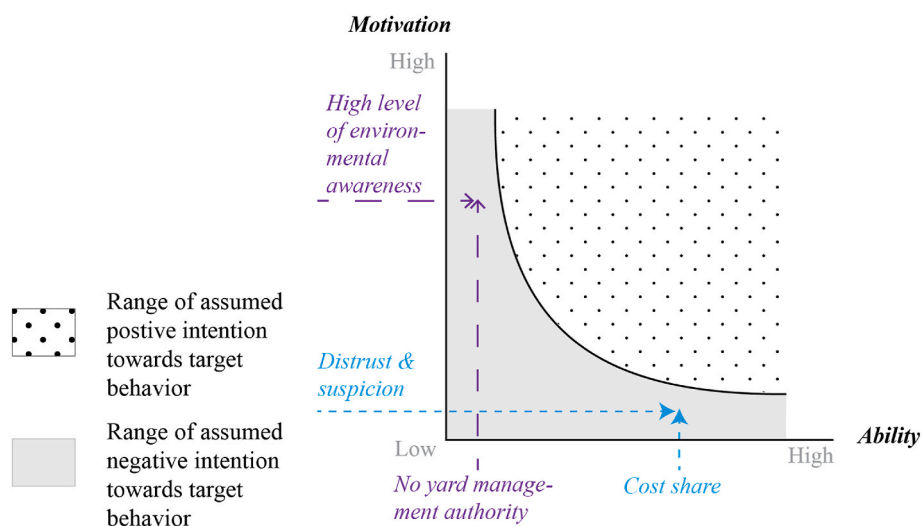
The case of rain garden adoption in Wisconsin, USA, provides another example illustrating how adoption is mediated by the interaction of ability and motivational factors (Belan and Nenn, 2010). Renters expressed interest in installing a rain garden after being informed about the environmental benefits of rain gardens via an outreach program; however, they did not have the authority to implement any yard changes (Fig. 5, Case Study B).

#### 4.3. Policy tools as prompts to address barriers to GSI adoption

Decision-makers are developing a wide range of planning policies and tools to facilitate the implementation of distributed GSI practices on private properties (Lieberherr and Green, 2018). Here we classify policy tools into (1) judicial control (sticks), (2) economic control (carrots), and (3) communication control (sermons) according to the typologies suggested by Bemelmans-Videc et al. (1998). Then, we draw parallels

between the three groups of policy instruments and the types of “prompts” in the Fogg Model (Fogg, 2009), which are (1) “spark” prompts for individuals who lack motivation, (2) “facilitator” prompts, which target motivated individuals who yet are reluctant to change their behavior because they perceive it as a complex action, and (3) “signal” prompts that work as a reminder for individuals who are motivated and capable (Fig. 6).

Our research uncovered two instances of judicial controls (i.e., sticks) for stormwater management at the parcel level, relevant to all three GSI practices discussed in this review. The first example is residential stormwater fees added to water bills in Ohio, USA. These fees, while a form of judicial control, are relatively low and, therefore, not a significant concern for households, indicating their limited effectiveness (Turner et al., 2016). From a municipal perspective, stormwater fees provide insufficient revenue to meet stormwater mandates or to cover retrofitting and maintenance costs (Cousins and Hill, 2021). Similarly, the second judicial enforcement, which involved limiting the impervious areas on individual residential parcels, did not lead to a sustained behavioral change for rain garden adoptions in North Carolina, USA, where inspections revealed that enforced households barely maintained their rain gardens (Woodward et al., 2009). To keep adopters from ending participation or to avoid poor maintenance over time, we suggest

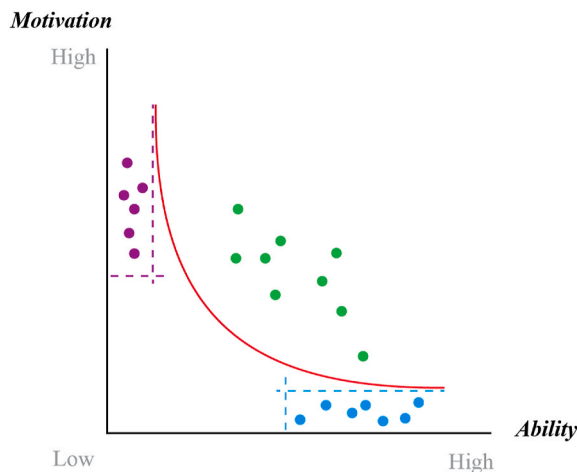


**Fig. 5.** An adapted Fogg Chart (Fogg, 2020). Non-adoption case studies.

Case study A (light blue arrows): non-adopters of rainwater tanks in Melbourne, Australia (Brown et al., 2016). The chart illustrates the positive influence of providing cost share through an auction process (positive ability factor) and the low motivation of residents due to distrust in the involved organization.

Case study B (purple arrows): non-adopters of rain gardens in Wisconsin, USA (Belan and Nenn, 2010). The chart illustrates the positive influence of an outreach addressing environmental awareness (positive motivation factor) and the lack of authority to implement yard changes on rental properties combining for a non-adoption decision.





**Fig. 6.** Adapted from [Guimaraes et al. \(2018\)](#): Fogg Chart. The figure illustrates how different types of prompts are required for the different adoption scenarios depending on the level of motivation and ability.

The red curve is the action line; individuals below this line have negative behavioral intentions, and those above have positive behavioral intentions.

The green dots represent individuals with positive intentions towards a target behavior because they have adequate levels of motivation and ability. This is where “**signal prompts**” are most effective.

The purple dots represent individuals with negative intentions towards a target behavior with relatively high motivation and low ability levels. The purple dotted line defines the area where a “**facilitator prompt**” would be required to achieve a behavioral change.

The blue dots represent individuals with negative intentions towards a target behavior with relatively high ability and low motivation levels. The blue dotted lines define the area where a “**spark prompt**” would be required to achieve a behavioral change.

that by-law obligatory tools be complemented by outreach programs meticulously addressing required maintenance effort, time, and cost. Additionally, in the case of rain gardens, expectations about aesthetics, i.e., seasonal alterations of plantings, are critical. A case in point is [Belan and Nenn's \(2010\)](#) recommendation for rain garden installation through planting instead of seeding to avoid extreme random growth patterns. Other complementary motivational factors supporting household compliance include education programs that incorporate both direct household benefits and the underlying environmental objectives.

Judicial control (i.e., sticks) could significantly influence a household's motivation to install a rain garden; however, they do not consider each resident's ability. In this context, we argue that open questions remain unanswered, such as whether these restrictive instruments are inclusive from a socioeconomic perspective ([Sun et al., 2024](#)) and to what extent such legal requirements would be socially acceptable and applicable ([Bicchieri, 2017](#)). In the context of residential GSI, such acceptability is reflected in alignment with the landscape norms within a community ([DeFields, 2013](#); [Nassauer et al., 2009](#); [Zmyslony and Gagnon, 2000](#)).

Stormwater credits and subsidized GSI programs are examples of economic control tools (i.e., carrots). In our review, case studies on such incentive programs targeted rain gardens and rainwater tanks or barrels, but none addressed permeable pavers. Unfortunately, municipal GSI programs demonstrate low household participation. For example, the overall uptake percentage by residents of the City of Kitchener, Ontario, for different GSI practices, including permeable pavers, was less than six percent of the population after five years of program initiation ([Aquafor Beech Ltd, 2016](#)). The evidence indicates that non-adopters persist for rain barrels and rain gardens despite subsidies, which aligns with more general notions about the limited effect of monetary tools in inducing sustainable behavioral change ([Bicchieri, 2017](#)).

Researchers noted the reluctance of households to participate in GSI

programs due to different considerations that vary according to GSI practice type, such as the unpleasant look of rain barrels, even if they were provided for free ([Meenar et al., 2022](#)), or for a hypothetical rebate program that lacked technical support for rain gardens ([Newburn and Alberini, 2016](#)). Distrust in GSI municipal programs also hindered the uptake of a cost-sharing program for rainwater tanks ([Brown et al., 2016](#)). In this context, we suggest citizen engagement would be a key strategy for building trust ([Green et al., 2012](#)). Therefore, municipal program managers should seek to establish robust cooperation with local bridging organizations and catalyze program familiarity through social learning ([Berkes, 2009](#)). Examples of such organizations are local environmental non-profits and neighbourhood associations having well-established connections with the local community ([Brown et al., 2016](#); [Chaffin et al., 2016](#)). Additionally, GSI program managers should explicitly state the contractual subsidy agreement, including data confidentiality measures, to clear suspicions of complicated application/enrollment processes ([Belan and Nenn, 2010](#)), and avoid upfront payments to mitigate household fraud concerns ([Brown et al., 2016](#)), if applicable.

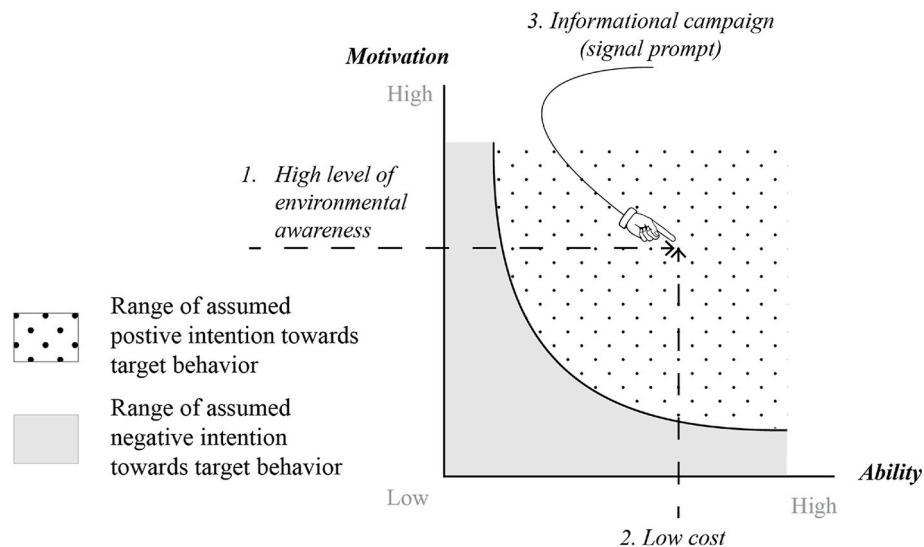
The last policy group used to support GSI uptake is communication controls (i.e., sermons), such as educational campaigns and outreach programs. Informational campaigns, in some cases, can act as a “signal” prompt or a reminder for pro-environmental residents who have the capability to install rain barrels ([Thurston et al., 2010](#)) (Fig. 7). However, being aware of rain barrels may be insufficient to catalyze persistent non-adopters, who also lack know-how as one of their top barriers ([Gao et al., 2016](#)). Furthermore, the complexities of maintaining rain barrels are a common factor for discontinuing their adoption ([Gao et al., 2016](#)).

These findings suggest that the duration of educational programs should be long enough to increase households' initial perceived capabilities and, in addition, provide technical support and guidance for adoption. Effective GSI installation and maintenance can be challenging even for professionals and may require context-specific standards that may vary from one intervention to another ([Zuniga-Teran et al., 2020](#)). Therefore, we suggest that one-on-one pre and post-adoption consultations could be required in some contexts. The City of Kitchener, Ontario, Canada, initiated a similar initiative to increase residential GSI uptake on private yards ([Reep Green Solutions, n.d.](#)). In collaboration with a local environmental non-profit, the Rain Smart Neighbourhoods program provides free on-site consultations where landscape professionals co-develop custom GSI interventions. The advisors offer technical advice on design, installation, and maintenance, and also recommend resources for required supplies and accessories. The program also includes follow-up surveys to support households beyond the initial adoption stage.

Beyond the need for technical training, more generally, previous research asserts that financial incentives are a requirement for outreach programs to achieve a higher uptake of pro-environmental behaviors ([Finger, 1994](#)). Furthermore, the impacts of formal environmental education and outreach programs on residents' perception of GSI adoption remain uncertain because the results of environmental outreach are inherently invisible, hard to quantify, and difficult to confirm ([Bemelmans-Videc et al., 1998](#)).

## 5. Conclusions

This review sheds light on the importance of recognizing differences in adoption factors for various GSI practices, and our findings call on scholars and practitioners to avoid using the generic term “green infrastructure” to generalize their findings. Rather, they should be clear about which GSI practice the results are relevant for and the targeted dependent variable. Focusing on individual GSI practices will help researchers and practitioners to comprehensively understand the unique challenges and opportunities associated with each intervention type. Policymakers, environmental planners, and stormwater managers can



**Fig. 7.** An adapted Fogg Chart (Fogg, 2009, 2020), for a case of rain barrel adopters in Ohio (USA). The chart illustrates the three catalysts of adoption (1) high motivation due to environmental awareness, (2) high financial capability due to the relatively low cost of rain barrels, and (3) informational campaign that acted as a “signal” prompt.

fine-tune and build on existing policy tools that cater to different GSI practices. Since each intervention type presents its specific challenges and benefits (Taguchi et al., 2020), a targeted policy framework is essential for promoting the widespread adoption of GSI practices and achieving transformative socio-ecological resilience within residential neighbourhoods.

Motivations are a critical yet unreliable determinant of GSI adoption. While egoistic motivations, such as direct household benefits, are essential for GSI adoption they fluctuate depending on several household attributes and contexts. On the other hand, given the inevitable heterogeneity of individual abilities, jurisdictional controls’ legitimacy and procedural fairness could be questioned. Further, municipal incentives and by-laws are insufficient due to constrained budgets and private property rights, respectively. From a policy perspective, neither voluntary GSI programs nor judicial measures to support GSI adoption might be efficient when applied separately. Instead, restrictive tools must be complemented with additional stimulative policy instruments such as educational campaigns (i.e., communication tools) that could facilitate and legitimize municipal enforcement. To confront the complexity of social systems and environmental uncertainties, environmental planners and policymakers should consider policy mixes that integrate carrots, sticks, and sermons for specific GSI practices (Pacheco-Vega, 2020) (Supplementary data 4). In this context, we suggest the Fogg model as a qualitative framework to guide the development of practice-specific integrated policy packages that account for households’ motivations and abilities.

## 6. Reflections for future research

We acknowledge limitations to this review that scholars could address in future research. Given that the Fogg Model is a qualitative framework, we do not discuss the magnitude with which each determinant influences households’ adoption decisions. Future works may consider conducting a meta-analysis for similar contexts to infer quantitative context-specific conclusions based on more in-depth statistical investigation. Additionally, we acknowledge that a few reviewed studies lacked statistical significance testing, potentially limiting our conclusions’ robustness.

We caution that our conclusions might have limited transferability across different built and natural environments. For instance, while social norms are acknowledged as a primary determinant of community

landscaping preferences and stewardship (Gobster et al., 2007; Nassauer, 2011), they vary across geographic and cultural contexts. We also suggest that future studies consider social desirability bias (Fleming and Savchenko, 2022), where responses by study participants reflect what residents perceive as socially acceptable answers instead of their actual preferences.

More empirical evidence on practice-specific determinants is required to develop evidence-based policy advice and address the challenge of cross-method and cross-location transferability of solutions (Cartwright and Hardie, 2012). Additionally, we highlight the need for more research on actual GSI programs that accurately capture revealed preferences and actions besides stated intentions for hypothetical scenarios. More specifically, future research could investigate more permeable pavers case studies to examine the impact of different policy tools on this understudied practice.

Beyond GSI, future research can explore nature-based solutions (NbS) as a broader strategy towards sustainable stormwater management. The three practices addressed in our review seek to conserve water resources by leveraging natural infiltration processes and can be considered types of NbS (FEMA, 2023; Santhanam and Majumdar, 2020). However, the distinction between using natural features (e.g., rain gardens) or engineered elements (e.g., permeable pavers and rain-water barrels or cisterns) raises interesting research questions for further investigation. For example, as people vary in their tendency to be ‘technophile’ or ‘ecophile’, do the adoption levels of various NbS practices vary between people when natural or engineered features are used to manage stormwater? Or do the motivational and ability factors influencing the adoption of various NbS practices vary between natural-based and engineered practices?

The quest for a comprehensive GSI framework that synchronizes land use, infrastructure, and social systems is yet to be completed. More systematic research is required to reach a comprehensive understanding of each system. In the context of social determinants of GSI adoption, scholars should address three levels of complexities: (1) identifying exploratory factors and their significance on the output variable, (2) synthesizing potential synergies and conflicts between these factors, and (3) quantifying the influence of local policy tools to address adoption barriers.

## CRediT authorship contribution statement

**Hazem Ahmed:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Dawn Cassandra Parker:** Writing – review & editing, Supervision, Resources, Methodology, Conceptualization. **Michael Drescher:** Writing – review & editing, Supervision, Conceptualization.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jenvman.2024.123279>.

## Data availability

Data will be made available on request.

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